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Personalized Language Learning Path Intelligent Recommendation: Algorithm Design Based on Collaborative Filtering and Knowledge Graph

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Abstract

In existing language learning platforms, personalized recommendation accuracy is insufficient, learning paths are rigid, and knowledge structures are ignored. This paper proposes to first construct a fine-grained language knowledge graph and use the TransR model to learn the complex semantic relationships between knowledge points; On this basis, a hierarchical attention collaborative filtering mechanism for knowledge perception is designed, which dynamically integrates user behavior preferences and knowledge structure information through dual attention at the project level and knowledge concept level; This article further introduces a path sequence generation strategy based on reinforcement learning, which drives agents to explore the optimal learning sequence in a knowledge graph environment using a multi-objective reward function. The average recommendation accuracy of the proposed algorithm is 71.2625%, the average path coverage is 64.9875%, and the knowledge structure utilization rate is 82.8%-88%, providing a feasible technical solution for precise recommendation in intelligent education systems.

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1 Introduction

With the rapid development of information technology and the continuous deepening of digital transformation in education, personalized learning has become the research core in the field of intelligent education. In the specific field of language learning, learners face unique challenges such as a vast knowledge system, diverse skill dimensions (vocabulary, grammar, pragmatics), and non-linear learning processes.

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This article defines ontology and relationship types applicable to language learning, and uses the TransR model to learn vector representations of knowledge points, thereby quantifying the complex semantic associations between them. It designs a dual layer attention network that, based on traditional collaborative filtering, not only focuses on the weights of user historical interaction items, but also further introduces neighborhood concept information of items in the knowledge graph, dynamically perceives users' potential knowledge states and weak links, and generates deeper user representations. This article formalizes path recommendation as a sequence decision problem in a knowledge graph environment, and designs a reward function that integrates real-time learning feedback, knowledge structure constraints, and long-term learning goals, enabling the system to automatically generate personalized, logical, and goal oriented learning path sequences through trial and error learning.

2 Related Work

The research on learning path recommendation is gradually integrating reinforcement learning, temporal analysis, and cognitive models to cope with the real-time changes in learners' states and the dynamic evolution of market demand. Kokkodis and Ipeirotis [1] proposed a reinforcement learning based career path recommendation framework, which takes market demand as environmental feedback and dynamically optimizes the order of skill acquisition through Q-learning algorithm. The existing adaptive learning systems rely on preset rules and cannot respond in real-time to changes in learners' cognitive states. Raj and Renumol [2] constructed a real-time learning analysis driven recommendation model, which dynamically adjusts the sequence of learning resources through an LSTM network. The Felder Silverman learning style model (such as visual/verbal, sequential/global) is widely used in e-learning, but lacks empirical verification of its recommendation effectiveness. Joseph et al. [3] designed a learning analysis intervention experiment to compare student learning behavior logs. Morozevich et al. [4] developed a path generation model based on XGBoost, which was piloted in IT enterprises. The model shortened the training cycle and improved the efficiency of employee project delivery. Benmesbah et al. [5] found that the convergence speed of improved GA is faster than that of standard GA, and the probability of generated paths violating constraints is reduced.

Elhossiny et al. [6] integrated AI and psychological models: using BigFive personality test to predict learning persistence, dynamically adjusting content difficulty, and using A* algorithm to search for the optimal path. Sun and Yuan [7] integrated graph embedding and attention mechanisms in a knowledge tracking model to design an adaptive learning path recommendation model (GE-MAKT) supported by deep learning. The research on learning path generation in e-Learning environment is scattered, and Ovtšarenko [8] conducted a systematic review. Varga et al. [9] used participatory action research to design peer mentoring systems, cultural identity courses, and track their academic retention rates. Benmesbah et al. [10] enhanced the real-time performance of GA by recalculating fitness based on the latest feedback from learners after each generation of evolution. This article proposes an algorithm design that integrates collaborative filtering and knowledge graph, aiming to capture group learning patterns through collaborative filtering, and use knowledge graph to reveal the internal structure and progressive relationship of language knowledge, thereby constructing a language learning path recommendation model that combines personalization and cognitive rationality, to promote the precise and intelligent development of intelligent education systems in the field of language learning.

3 Method

3.1. Construction of Fine-Grained Language Knowledge Graph

The knowledge graph primarily defines the ontology by instantiating key concepts in the field of language learning as entities, such as "present perfect tense", "business English email beginning", "HSK CET-4 vocabulary - economy", etc. At the same time, multiple semantic relationship types between entities are clearly defined, mainly including "premise relationship" (B must be mastered before learning A), "component relationship" (knowledge point A belongs to course unit B), and "correlation relationship" (knowledge point A and knowledge point B are highly related in semantics or usage scenarios). Based on this ontology, extract triplet facts from structured course outlines, textbook catalogs, and expert experience [11-12]. Subsequently, the TransR model was used for embedding

learning on this graph. TransR maps entities and relationships to different semantic spaces for computation; For each triplet (h, r, t) , where h and t are the vectors of the head and tail entities, and r is the relationship vector, TransR first projects the entity vector onto the semantic space of relationship r through a relationship specific projection matrix M_r , obtaining the projected vector:

$$h_{\perp} = hM_r \quad (1)$$

$$t_{\perp} = tM_r \quad (2)$$

$h_{\perp}$ and $t_{\perp}$ are the projection vectors of the head entity and tail entity in the relation r space, respectively.

Then, in the relational space, by learning to associate the head and tail entity vectors projected from the triplets that satisfy the relationship with r , the scoring function is:

$$f_r(h, t) = \|h_{\perp} + r - t_{\perp}\|_2^2 \quad (3)$$

The model optimizes embedding by minimizing this score. This process brings entities with the same relationship closer together in a relationship specific vector space, capturing the complex and asymmetric dependencies and associations between language knowledge points, and providing high-quality, computable knowledge representations for subsequent recommendations.

3.2. Hierarchical Attention Collaborative Filtering Mechanism for Knowledge Perception

Traditional collaborative filtering only utilizes the user item interaction matrix, which suffers from sparsity and cold start issues. This mechanism enhances its depth by introducing knowledge graph embedding. The model input includes the user's historical interaction sequence and the embedding vectors of all items in the knowledge graph. Firstly, the initial ID embeddings of users and projects are obtained through an embedding layer [13-14]. The key innovation lies in the design of hierarchical attention networks. The first layer is project level attention, which analyzes the sequence of projects that the user has interacted with in history, assigns different weights to each historical project, distinguishes the user's long-term interests from recent attention points, and mitigates the impact of noisy interactions. It is represented as calculating the attention weight of each item in the historical project sequence:

$$\alpha_i = \text{softmax}(v_a^T \cdot \tanh(W_a e_i + b_a)) \quad (4)$$

e_i is the embedding vector of project i , W_a and b_a are learnable parameters, v_a^T is the weight vector, and α_i is the normalized attention score of project i . The second layer is knowledge concept level attention, which further introduces neighboring information such as prerequisite knowledge points and related knowledge points of the project in the knowledge graph on the basis of project level attention. The model calculates the correlation between user representations and neighboring knowledge point representations, thereby dynamically perceiving potential, unexpressed knowledge states or weak links of the user. For each neighbor concept c of project i , its correlation score is:

$$s_c = u^T \cdot e_c \quad (5)$$

u is the current user representation vector, e_c is the knowledge graph embedding vector of concept c , and T is the transpose operation. Ultimately, the overall representation of the user is the fusion of their ID embedding u_{id} , weighted historical interaction item embedding $\sum_i \alpha_i e_i$, and activated related knowledge concept embedding $\sum_c \text{softmax}(s_c) e_c$, which is:

$$u_{\text{final}} = u_{id} + \sum_i \alpha_i e_i + \sum_c \text{softmax}(s_c) e_c \quad (6)$$

This design enables the model to not only rely on co-occurrence behavior, but also infer the user's deep learning state through knowledge graph inference.

3.3. Path Sequence Generation Strategy Based on Reinforcement Learning

After obtaining the deep representation of the user, path generation is formalized as a sequence decision problem, which is interactively learned by an intelligent agent (recommendation system) in an environment composed of knowledge graphs. The state is defined as the encoding of the dynamic representation of the current user and the set of learned knowledge points [15-16]. The action space refers to all the recommended knowledge point entities in the knowledge graph. The design of the reward function balances multiple objectives: real-time rewards are based on user feedback after learning recommended knowledge points (such as test accuracy); Mid term rewards encourage intelligent agents to recommend actions that conform to the constraints of the knowledge graph structure, such as prioritizing the recommendation of knowledge points that have met all prerequisite conditions and are highly relevant to the current state. Long term rewards are linked to the overall achievement of users' learning goals, such as mastering all core concepts of a course module. The strategy network adopts the Actor Critic architecture, where the Actor network outputs the probability distribution of actions (knowledge points) based on the current state, and the Critic network evaluates the long-term value expectation of the state.

Table 1 shows the configuration of key parameters for the path generation strategy network:

Table 1. Key parameter configuration of path generation strategy network

Parameter Category	Specific Parameter	Parameter Value / Setting
Network Architecture Parameters	Actor Network Hidden Layers	[256, 128]
	Critic Network Hidden Layers	[128, 64]
	Activation Function	ReLU
Training Parameters	Learning Rate (Actor)	0.0001
	Learning Rate (Critic)	0.001
	Discount Factor (γ)	0.95
	Experience Replay Buffer Size	50000
	Initial Exploration Rate (ϵ)	0.5
Environment & Exploration Parameters	Exploration Rate Decay	0.995
	Minimum Exploration Rate	0.01
	Immediate Reward Weight (α)	1.0
Reward Function Weights	Structural Reward Weight (β)	0.3
	Long-term Goal Reward Weight (γ)	0.5

During the training process, the agent continuously optimizes its strategy by simulating trial and error, with the goal of updating parameters by maximizing the expected cumulative reward:

$$J(\theta) = E_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t, s_{t+1}) \right] \quad (7)$$

The strategy with τ as the trajectory, π_{θ} as the parameter θ , γ as the discount factor, and R as the reward function is used to calculate the immediate reward value obtained when transitioning from state s_t to state s_{t+1} after performing action a_t .

This ultimately learns to automatically generate personalized learning path sequences that are adaptable to individual knowledge gaps, follow the internal logic of the discipline, and can be dynamically adjusted to maintain learning motivation.

4 Results and Discussion

4.1. Experimental Setup

This work has used the open language learning dataset Lang-8 for the evaluation benchmark. This dataset has large amount of real user text exercises and its corresponding grammar knowledge point annotations. Systematic preprocessing was conducted on the raw data through the error labeling cleaning, knowledge point standardization alignment and sequence construction steps, and the resulting user knowledge point interaction sequences are of high quality. The data is divided into training set, validation set, and testing set in the ratio 8:1:1 in chronological order to simulation real progressive learning scenarios. In order to thoroughly test the performance of the algorithm proposed

in this paper, four representative baseline models were chosen for comparison, including traditional collaborative filtering algorithm based on group behavior pattern (ItemCF), knowledge graph based recommendation algorithm based on only knowledge structure association (KG Only), rule-based recommendation algorithm based on teaching expert experience to set fixed order (Rule-Based), and content-based recommendation algorithm based on learning resource content similarity (Content-Based).

4.2. Result Analysis

1) Recommended performance for cold start

The recommendation performance was systematically evaluated through accuracy, and the results are shown in Figure 1:

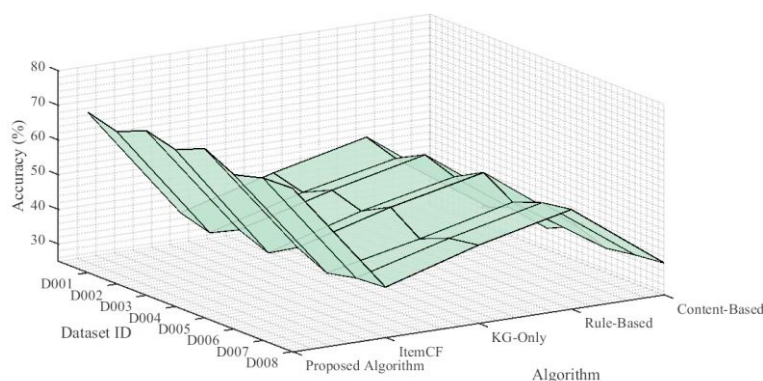


Fig 1. Recommended accuracy

This algorithm demonstrates significant and stable performance advantages. The data shows that its accuracy fluctuates high within the range of 68.9% to 73.8%, with an average of 71.2625%, significantly higher than other baseline models. In contrast, the second best performing Rule-Based accuracy ranges from 48.9% to 54.6%, with an average of 51.75%; However, ItemCF and Content-Based performed the weakest, with average accuracies of only 38.35% and 32.875%, respectively. This result clearly demonstrates that by introducing knowledge graph embedding and hierarchical attention mechanism, this algorithm effectively overcomes the limitations of traditional methods in sparse user interaction data, achieves collaborative utilization of group learning rules and inherent knowledge structure, and can still make accurate recommendations in the absence of sufficient historical behavior.

2) Evaluation of Learning Path Generation Effectiveness

This article compares the performance of five recommendation algorithms in path coverage, and the results are shown in Figure 2.

The highest average value of 64.9875% in path coverage was achieved by an algorithm in this article, which shows stable performance to all sets of test samples (61.4% -68.9%). The performance of ItemCF and Content-Based recommendations proves the weakest and the values are found to be 40.0125% and 36.025% respectively. This demonstrates that by relying only on behavioral co-occurrence or content similarity, immutability of the recommendation paths can hardly be guaranteed and complete. Although the recommendation based on knowledge graph (47.7375%) and Rule-Based (53.55%) have improved due to the introduction of prior knowledge structure, they are still significantly lower than the algorithm proposed in this paper. The analysis shows that the high average coverage of algorithm in this article is due to the effective combination of collaborative filtering and knowledge graph integration, collaborative filtering mines the learning patterns in groups, and in turn, it has a wide range of coverage. The knowledge graph ensures a logical coherence and the fulfillment of prerequisites between recommended knowledge points so that the guarantee of coverage depth is ensured. This dual mechanism allows the algorithm to more adequately reach potential knowledge states of learners and thus creates a wide and consistent path for learning.

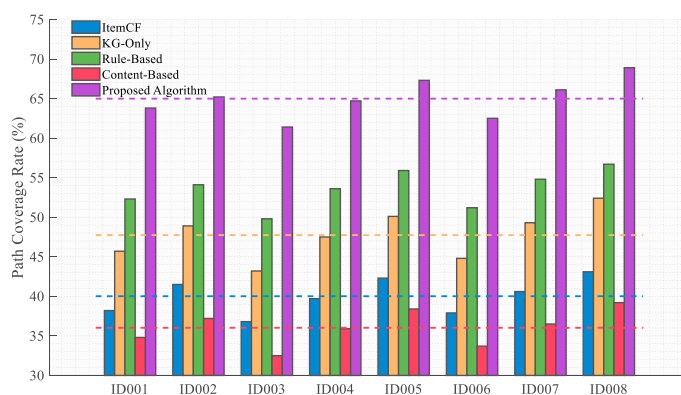


Fig 2. Path Coverage

3) Knowledge Structure Utilization

To quantitatively evaluate the degree of compliance of different algorithms with predefined knowledge graph structures, the proportion of continuous knowledge point pairs that conform to strong logical associations such as "premise relationships" and "components" in the knowledge graph is calculated in the recommendation path to measure the knowledge coherence and structure of the recommendation path. The results are shown in Figure 3:

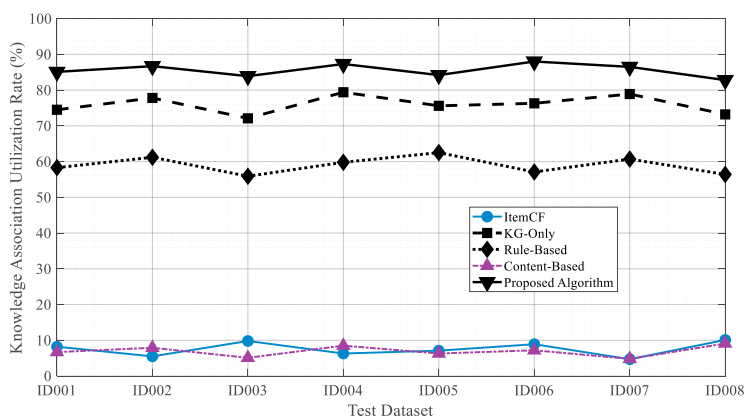


Fig 3. Knowledge Structure Utilization Performance

This algorithm not only has the highest overall level, but also has a very small fluctuation range, demonstrating excellent robustness (82.8% -88%). This stability stems from the dynamic perception of users' deep states by the hierarchical attention mechanism and the continuous optimization of structural constraints by reinforcement learning strategies, enabling the algorithm to calmly cope with the challenges brought by different test sets. Although the KG Only algorithm ranks second overall, it shows a decline on datasets such as ID003 and ID008, reflecting the sensitivity of pure structure recommendation to specific local structures or test sample distributions in knowledge graphs. Its performance is easily affected by the connection density of the graph itself or the coverage range of the test set. Rule-Based performs steadily but with limited levels, confirming the ceiling of fixed rule scalability. The recommendations of ItemCF and Content-Based always fluctuate violently and randomly at low levels, which once again emphasizes that recommendations that are not guided by knowledge structures are unpredictable and unreliable in terms of coherence.

4) Ablation Experiment

This article removes or replaces key modules (hierarchical attention HAN, reinforcement learning strategy RL, knowledge graph constraint KG) one by one, retrains and evaluates the comparative model, calculates accuracy, path coverage, and structural utilization, and the results are shown in Table 2:

Table 2. Contribution analysis data of each component in the ablation experiment

Model Variant	Recommendation Accuracy (%)	Path Coverage (%)	Knowledge Structural Utilization (%)
Full Model (Ours)	71.3	64.2	85.4
w/o Hierarchical Attention Network (HAN)	64.7	58.1	79.8
w/o RL Module (replaced with greedy selection)	68.5	57.3	81.6
w/o Knowledge Graph Constraints (KG)	65.9	60.5	52.7
w/o HAN & KG	58.2	51.4	48.9

The complete model performs the best in all three indicators, verifying the effectiveness of the fusion design of CF, knowledge graph, and reinforcement learning. Removing the hierarchical attention mechanism (w/o HAN) resulted in a significant decrease in accuracy (-6.6%), with synchronous declines in coverage and structural utilization, indicating that it also helps to improve the breadth and logicity of recommendations. When using greedy selection strategy instead of reinforcement learning module (w/o RL), the path coverage decreases the most significantly (-6.9%), and there is also a loss in accuracy and structural utilization. This confirms the key role of reinforcement learning in exploring long-term optimal sequences, avoiding local optima, and improving path diversity and integrity. Removing the knowledge graph constraint (w/o KG) alone resulted in a cliff like drop in knowledge structure utilization to 52.7%, and accuracy and coverage were also affected, highlighting the irreplaceability of knowledge graphs in ensuring logical coherence of learning paths.

5 Conclusion

This research algorithm can well improve the effective factors of accuracy, coverage of the path and knowledge coherence of the corresponding recommendations, verify the effect of group learning rules plus domain knowledge structures, and reinforce learning sequence optimization and effective. However, there are still some limitations of this study, such as that the performance of this algorithm is dependent on the quality and completeness of knowledge graph construction, and the impact of non-cognitive parameters such as emotional states of learners on their learning paths is not considered sufficiently. Future work can be directed in constructing finer grained, cross lingual knowledge graphs towards a more humanized and adaptive next generation intelligent language learning.

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